19 $y = \dfrac{3}{5k + 3}$ and $k = \dfrac{3}{5x - 3}$

Find an expression for y in terms of x.
Give your answer in its simplest form.

Substitute k into y = 3/(5k + 3).

$$\frac{3}{5[3/(5x - 3)] + 3}$$

1 mark
Replace k by the whole fraction.

$$5k + 3 = 15/(5x - 3) + 3$$

$$\frac{15 + 3(5x - 3)}{5x - 3} = \frac{15x + 6}{5x - 3}$$

1 mark
Use a common denominator.

Therefore y = 3(5x - 3)/(15x + 6)

1 mark
Divide by a fraction by multiplying by its reciprocal.

$$\frac{5x - 3}{5x + 2}$$

1 mark
Cancel the common factor 3.

Original expressions require x ≠ 3/5 and x ≠ −2/5.

(Total for Question 19 is 4 marks)

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